

Department of Computer Science

Gujarat University

DSC-M-CS-314T Data Analytics (Practical)

Assignment - Unit 1

1. Create a data frame of the below table.

Name	Age	Salary
Mahesh	23	10000
Kiran	25	12000
Suresh	28	15000
Sumit	26	14000
Ramesh	27	25000
Sawpnil	30	20000
Swati	29	26000

2. Get the maximum values of salary from the data frame.
3. Get the minimum values of the salary from the data frame.
4. Write the statement which will sort the marks in the Data Frame, in descending order.
5. Write a program to plot the ogive of random data from 1 to 20.
6. Write a program to plot the area chart of the data from 1 to 30.
7. Create the Data Frame of the below table.

Name	Maths	English	Science
Ramesh	78	67	56
Vedika	76	75	47
Harun	84	59	60
Prasad	67	72	54

8. Access the element in the 1st row in the 3rd column.
9. Access all the element in 3rd column.
10. Access elements of 2nd and 3rd row from 1st and 2nd column.
11. Write a program to plot a pie chart of values=[3,4,7,9] and categories=['A', 'B', 'C', 'D'].
12. Write the code for outlier removal from scratch.
13. [4, 7, 10, 14, 36, 16, 18, 20, 67, 22, 2]
14. Write the code of Finding Covariance from given data.
15. X= [2, 4, 6, 8, 10] , Y= [1, 3, 5, 7, 9]
16. Write a Python program from scratch to calculate the mean of [10, 20, 30, 40, 50].
17. Using NumPy, calculate the mean and median of [12, 15, 18, 20, 25, 30].
18. Write a Python program from scratch to find the median of [15, 10, 20, 25, 30].

19. Using Pandas, calculate the mean, median, and mode of a DataFrame column named Income.
20. Write a Python program from scratch to find the mode of [10, 20, 20, 30, 40, 20, 50].
21. Using NumPy, calculate the range of [15, 20, 25, 30, 45, 50].
22. Write a Python program from scratch to calculate the population variance of [10, 12, 15, 18, 20].
23. Using NumPy, calculate the variance and standard deviation of [10, 12, 15, 18, 20].
24. Write a Python program from scratch to calculate the covariance between $X = [10, 20, 30, 40, 50]$ and $Y = [15, 25, 35, 45, 55]$.
25. Using Pandas, calculate the correlation coefficient between Income and Loan_Amount.
26. Write a Python program from scratch to calculate Pearson's correlation coefficient for two numerical lists.
27. Using Matplotlib, create a scatter plot to show the relationship between Income and Loan_Amount.
28. Use Python's random library to simulate 1,000 coin tosses and calculate the experimental probability of getting Heads.
29. Use Python's random library to simulate 10,000 die rolls and calculate the probability of obtaining a 6.
30. Write a Python program to demonstrate the complement rule using a coin-toss experiment.
31. Using Python collections, count the occurrence of each outcome in ['H','T','H','H','T','T','H','T'] and calculate their probabilities.
32. Write a Python program to verify that $P(A) + P(A^c) = 1$ using a random experiment.
33. Simulate two independent coin tosses and calculate the experimental probability of obtaining two Heads.
34. Generate 1,000 random observations from a Binomial distribution using NumPy with $n=10$ and $p=0.5$.
35. Generate 1,000 random observations from a Poisson distribution with $\lambda = 4$ using NumPy.
36. Generate 1,000 observations from a Normal distribution with mean 50 and standard deviation 10 using NumPy.
37. Generate standard normal random observations using NumPy and calculate their mean and standard deviation.
38. Using Python, calculate the Binomial probability of exactly 2 successes in 5 trials when $p = 0.4$.

39. Using SciPy, calculate the probability of exactly 3 events occurring when $\lambda = 5$ in a Poisson distribution.
40. Using SciPy, calculate the probability that a normally distributed observation with mean 100 and standard deviation 15 is greater than 120.
41. Create a DataFrame containing Income, Loan_Amount, and Credit_Score. Using Pandas, generate a correlation matrix and identify the strongest relationship.
42. Write a Python program from scratch to calculate Pearson correlation for $X = [2, 4, 6, 8, 10]$ and $Y = [5, 8, 12, 15, 20]$. Do not use `np.corrcoef()`.
43. Using Pandas and Matplotlib, calculate the covariance and correlation between customer income and loan amount and display their relationship using a scatter plot.
44. Use Python's random library to simulate 10,000 coin tosses. Calculate the probabilities of Heads and Tails and verify the complement rule.
45. A box contains 5 red, 3 blue, and 2 green balls. Use Python to simulate 10,000 random selections and calculate the experimental probability of Red, Blue, Green, and Not Red.
46. Simulate 10,000 die rolls using Python and demonstrate the addition law of probability for the events: A = getting an even number and B = getting a number greater than 3.
47. A dataset contains Gender and Loan_Approved. Using Pandas, calculate $P(\text{Loan Approved})$, $P(\text{Female})$, and $P(\text{Loan Approved} \mid \text{Female})$ using a contingency table.
48. Simulate two dice being rolled 10,000 times and verify the multiplication law for independent events using the events A = first die is 6 and B = second die is 6.
49. Write a Python program from scratch to calculate Binomial probabilities for $n = 10$ and $p = 0.3$ for $X = 0, 1, 2, 3$, and 4.
50. Using SciPy, perform a complete Binomial distribution analysis for $n = 20$ and $p = 0.4$. Calculate $P(X=5)$, $P(X \leq 5)$, $P(X \geq 5)$, mean, and variance.
51. A bank receives an average of 4 complaints per hour. Write a Python program from scratch using the Poisson formula to calculate $P(X=0)$ through $P(X=6)$.
52. Using SciPy, perform a Poisson distribution analysis for $\lambda = 3$. Calculate $P(X=0)$, $P(X=2)$, $P(X=5)$, $P(X \leq 3)$, and $P(X > 3)$, and plot the distribution.
53. A population contains 20 transactions, of which 5 are fraudulent. Four transactions are selected without replacement. Using SciPy's hypergeometric distribution, calculate the probability of exactly 2 fraudulent transactions and at least 1 fraudulent transaction.
54. Write a Python program from scratch to calculate z-scores for $X = 30, 40, 50, 60$, and 70 when the mean is 50 and standard deviation is 10. Interpret the results.

55. Transaction processing time is normally distributed with mean 10 minutes and standard deviation 2 minutes. Using SciPy, calculate $P(X < 8)$, $P(X > 12)$, and $P(8 < X < 12)$, and interpret the results.
56. Generate 10,000 observations from a normal distribution with mean 100 and standard deviation 15 using NumPy. Calculate the sample mean and standard deviation and display the distribution using a histogram.

(Descriptive Statistics)

(Book name- “Statistics for Business & Economics” by Anderson)

Note: using list, data-frame and csv file and array– (all Exercise Example)

Title	Content	Book Chapter	Example No.
Descriptive statistics	Measures of location, Measures of variability, Measures of association between two variables, Measures of distribution	3	Cover all topics: write 5 examples of each topic
Probability	basic relationships of probability	4	24,25,27,30,31,32
Probability distribution	Binomial, Poisson, Hyper geometric, Normal and standard normal	5 and 6	Chapter 5 – 28,29,31,32,33,40,41,42,49,50,51 Chapter 6- cover all topics and write 10 examples each topic